SIMATS ENGINEERING
ASSESSMENT TOOL3

COURSE CODE: MLA 02 **COURSE NAME: Fundamentals Of Machine Learning**

COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.* *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I **R.Pranathi(192525076)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Pranathi

Annexure A

Debugging & Optimisation Task

Q1. A real estate company developed a linear regression model to predict house prices, but the model is showing high prediction error.

(i) Identify possible causes for the high error (e.g., multicollinearity, outliers, poor feature selection).

(ii) Analyze the dataset/model behavior to detect these issues.

(iii) Apply suitable optimization techniques (such as feature scaling, removing outliers, or adding relevant features) to improve the model performance. (5 Marks-8 Mins)

Q2.An email classification system using a linear model is incorrectly classifying several spam emails as non-spam.

(i) Identify the reasons for misclassification (e.g., improper decision boundary, imbalanced data).

(ii) Examine the current decision boundary and classification errors.

(iii) Refine the model using optimization techniques such as adjusting thresholds, feature engineering, or re-training with balanced data.(5 Marks-8 Mins)

Q3.A Naïve Bayes classifier is giving poor accuracy due to missing or zero probability values in the dataset.

(i) Identify the issue related to missing or zero probabilities.

(ii) Analyze how these values affect classification results.

(iii) Apply appropriate debugging and optimization methods such as Laplace smoothing to improve accuracy. (5 Marks-8 Mins)

Q4.A discriminant function is unable to clearly separate two classes, leading to incorrect classifications.

(i) Identify possible reasons for poor class separation (e.g., overlapping features, incorrect weights).

(ii) Analyze the decision boundary and feature contribution.

(iii) Improve the discriminant function using optimization strategies such as adjusting weights, feature transformation, or normalization.(5 Marks-8 Mins)

Q5.A Bayesian logistic regression model developed for customer churn prediction is suffering from overfitting, resulting in poor generalization.

(i) Identify signs and causes of overfitting in the model.

(ii) Analyze model performance on training vs testing data.

(iii) Apply debugging and regularization techniques (such as L1/L2 regularization or cross-validation) to improve model performance.(5 Marks-8 Mins)

Q6.A linear regression model built for predicting sales is performing poorly because the features (e.g., advertising budget in thousands and number of outlets in units) are on different scales.

(i) Identify how feature scaling issues affect model performance.

(ii) Analyze the impact of unscaled features on regression coefficients.

(iii) Apply appropriate optimization techniques such as normalization or standardization to improve the model.(5 Marks-8 Mins)

Q7.A binary classification model for fraud detection is biased towards the majority class, leading to poor detection of fraudulent cases.

(i) Identify the problem caused by class imbalance.

(ii) Evaluate model performance using suitable metrics (precision, recall, F1-score).

(iii) Apply optimization techniques such as resampling (oversampling/undersampling) or class weighting to improve performance.(5 Marks-8 Mins)

Q8.A Naïve Bayes classifier is underperforming because some input features are highly correlated, violating the independence assumption.

(i) Identify how correlated features affect Naïve Bayes performance.

(ii) Analyze feature relationships in the dataset.

(iii) Apply optimization techniques such as feature selection or dimensionality reduction to improve accuracy. (5 Marks-8 Mins)

Q9.A logistic regression model is unable to clearly separate two classes due to overlapping feature distributions.

(i) Identify the cause of overlapping decision boundaries.

(ii) Analyze the feature space and classification errors.

(iii) Improve the model using techniques such as feature transformation, polynomial features, or regularization. (5 Marks-8 Mins)

Q10.A discriminant analysis model is too simple and fails to capture the complexity of the dataset, resulting in low accuracy.

(i) Identify the signs of underfitting in the model.

(ii) Analyze model bias and feature limitations.

(iii) Apply optimization strategies such as adding features, increasing model complexity, or using nonlinear transformations. (5 Marks-8 Mins)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1ANS:

(i) Identify the possible causes of high prediction error

Linear regression is a supervised machine learning algorithm used to predict continuous numerical values. High prediction error occurs when the model cannot accurately learn the relationship between house features and prices.

The possible causes are:

1. **Multicollinearity:** Two or more input features, such as house area and number of bedrooms, may be highly correlated. This can make regression coefficients unstable.
2. **Outliers:** Houses with unusually high or low prices may influence the regression line and increase prediction errors.
3. **Poor feature selection:** Important features, such as location, house age, and building condition, may be missing.
4. **Missing or incorrect data:** Incomplete records and incorrect prices can reduce model accuracy.
5. **Underfitting:** A simple linear model may fail to capture complex or nonlinear relationships between house features and prices.

(ii) Analyze the dataset and model behaviour

The following methods can help identify the problems:

- **Correlation analysis:** Check whether input features are strongly correlated.
- **Scatter plots:** Plot house area against price to examine their relationship.
- **Box plots:** Identify unusually high or low values.
- **Residual analysis:** Calculate the difference between actual and predicted prices.
- **Error evaluation:** Calculate Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

The residual is calculated as:

$$e_i = y_i - \hat{y}_i$$

Where:

- y_i= Actual house price.
- $\hat{y}_i$= Predicted house price.
- e_i = Prediction error.

Large residuals indicate that the model has made inaccurate predictions for certain houses.

(iii) Apply suitable optimization techniques

1. Feature scaling: Standardize numerical features when appropriate to improve the numerical stability of optimization algorithms.

2. Outlier handling: Investigate unusual observations and correct data-entry errors. Genuine expensive houses should not be removed without justification.

3. Feature engineering: Add important features such as location, house age, number of bathrooms, and building condition.

4. Feature selection: Remove redundant or irrelevant features to improve model stability.

5. Cross-validation: Evaluate the model using multiple training and validation splits to estimate its generalization performance.

6. Nonlinear modelling: If the relationship between house features and prices is nonlinear, consider polynomial regression or another suitable model.

Conclusion

By identifying outliers, improving feature selection, correcting data problems, and selecting an appropriate model, the company can reduce prediction errors and improve house price prediction.

2ANS:

(i) Identify the reasons for misclassification

Linear classification is used to divide observations into different classes using a decision boundary.

The possible reasons for incorrect spam classification are:

1. **Improper decision boundary:** The model may not have learned an effective separation between spam and non-spam emails.
2. **Imbalanced dataset:** The training dataset may contain significantly more non-spam emails than spam emails.
3. **Insufficient features:** Important information, such as suspicious links and promotional words, may be missing.
4. **Incorrect labels:** Some emails may have been labelled incorrectly during training.
5. **Poor model training:** The model may not have learned the important patterns in spam emails.

(ii) Examine the decision boundary and classification errors

A linear classifier can be represented as:

$$Z = \omega_0 + \omega_{1_1} x_1 + w_2 x_2$$

Where:

- x_1= Frequency of suspicious words.
- x_2= Frequency of promotional words.
- $\omega_0, \omega_{1,} \omega_2$= Model parameters.

The decision boundary is:

$$\omega_0 + \omega_1 x_1 + \omega_2 x_2 = 0$$

The classifier assigns emails to classes according to which side of the boundary they fall on.

A confusion matrix can identify four types of predictions:

- True Positive: Spam correctly identified as spam.
- True Negative: Non-spam correctly identified as non-spam.
- False Positive: Non-spam incorrectly classified as spam.
- False Negative: Spam incorrectly classified as non-spam.

False negatives are particularly important in this scenario because they represent spam emails that reach the inbox.

(iii) Refine the model using optimization techniques

1. Adjust the classification threshold: Lowering the threshold for predicting spam can increase spam detection, although it may also increase false positives.

2. Feature engineering: Add features such as suspicious URLs, sender information, and the frequency of spam-related words.

3. Balance the training dataset: Use oversampling or undersampling to reduce the effect of class imbalance.

4. Correct the training labels: Identify incorrectly labelled emails and correct them.

5. Retrain the model: Train the classifier using improved features and representative data.

6. Evaluate the results: Measure precision, recall, and F1-score to determine whether spam detection has improved.

Conclusion

By adjusting the decision threshold, improving input features, and handling class imbalance, the email service can reduce false negatives and identify more spam emails accurately.

3ANS:

(i) Identify the issue related to missing or zero probabilities

Naïve Bayes is a probabilistic classification algorithm based on Bayes' theorem.

Its formula is:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

Where:

- P(C|X) = Probability of class CC given the features.
- P(X|C) = Probability of the features given the class.
- P(C) = Prior probability of the class.
- P(X) = Probability of the observed features.

A zero probability occurs when a particular feature value does not appear in the training observations belonging to a class.

For example, if the word "discount" never appears in the spam training data, its estimated probability given the spam class may be zero.

Missing data can also affect probability estimation if it is not handled correctly.

(ii) Analyze how zero probabilities affect classification

Naïve Bayes calculates class probabilities by multiplying conditional probabilities.

For example:

$$P(\text{Spam} \mid X) \propto P(\text{Spam})\ P(\text{Offer} \mid \text{Spam}) P(\text{Win} \mid \text{Spam})$$

Suppose:

$$P(\text{Win} \mid \text{Spam}) = 0$$

Then:

$$P(\text{Spam} \mid X) \propto 0$$

The estimated probability of spam becomes zero, even if the other features strongly indicate that the email is spam.

As a result, the classifier may assign the email to the wrong class.

(iii) Apply debugging and optimization techniques

1. Laplace smoothing

Laplace smoothing is used to avoid zero probabilities by adding a small positive value to the observed counts.

For categorical features, the formula is:

$$p(x_i|C) = \frac{N_{ic}+\alpha}{N_C+\alpha k}$$

Where:

- N_{iC} = Count of the feature value in class CC.
- Nc= Total number of observations in class CC.
- K = Number of possible feature values.
- α = Smoothing parameter, commonly set to 1.

2. Handle missing values: Investigate why values are missing and use an appropriate imputation or missing-data strategy.

3. Clean the dataset: Remove duplicate records and correct inconsistent data.

4. Select useful features: Remove irrelevant features that contribute little to classification.

5. Evaluate the model: Test the classifier on unseen data and compare accuracy, precision, recall, and F1-score before and after smoothing.

Conclusion

Zero probabilities can cause Naïve Bayes to produce incorrect class estimates. Laplace smoothing and proper data preprocessing help prevent this problem and improve classification reliability.

4ANS:

(i) Identify the possible reasons for poor class separation

A discriminant function is used to distinguish between classes based on their input features.

The possible reasons for poor separation are:

1. **Overlapping features:** Both classes may contain similar feature values.
2. **Incorrect weights:** The coefficients may not represent the differences between classes accurately.
3. **Irrelevant features:** Unnecessary input features may introduce noise.
4. **Different feature scales:** Features measured using different units may affect numerical estimation.
5. **Incorrect model assumptions:** A linear discriminant function may not be suitable when the classes have complex nonlinear boundaries.

(ii) Analyze the decision boundary and feature contribution

A linear discriminant function is represented as:

$$g(X) = \omega_0 + \omega_1 x_1 + \omega_2 x_2$$

Where:

- x_1, x_2= Input features.
- ω_0= Intercept.
- ω_1, ω_2= Feature coefficients.

The decision boundary is:

$$\omega_0 + \omega_1 x_1 + \omega_2 x_2 = 0$$

The model assigns observations to different classes depending on the value of the discriminant function.

To analyze the problem:

1. Plot the observations belonging to both classes.
2. Identify regions where the classes overlap.
3. Examine the coefficients to understand feature contributions.
4. Use a confusion matrix to identify incorrect classifications.
5. Check whether the model assumptions match the dataset.

(iii) Improve the discriminant function using optimization strategies

1. Adjust the weights: Re-estimate the coefficients using appropriate training data.

2. Normalize or standardize features: Bring numerical features to comparable scales where appropriate.

3. Perform feature selection: Remove irrelevant or redundant variables.

4. Apply feature transformation: Transform the features to improve the separation between classes.

5. Use a more suitable model: If a linear decision boundary is insufficient, consider a nonlinear classifier or Quadratic Discriminant Analysis when its assumptions are appropriate.

6. Evaluate performance: Compare the original and improved models using accuracy, precision, recall, and a confusion matrix.

Conclusion

Analyzing feature contributions, improving data quality, and selecting a suitable discriminant function can improve class separation and reduce classification errors.

5ANS:

(i) Identify the signs and causes of overfitting

Bayesian logistic regression estimates the probability that a customer will discontinue a service. Overfitting occurs when the model learns the training data too closely, including noise and accidental patterns.

The main signs and causes are:

1. **High training accuracy:** The model performs extremely well on training data.
2. **Low testing accuracy:** Its predictions are less accurate on unseen customers.
3. **Excessive model complexity:** Too many features may be used relative to the amount of training data.
4. **Noisy data:** Incorrect labels and irrelevant variables may affect the learned relationships.
5. **Large coefficients:** Some coefficients may become excessively large when the data do not adequately constrain them.

(ii) Analyze training and testing performance

The model should be evaluated using separate training and testing datasets.

For example:

Dataset	**Accuracy**
Training data	98%
Testing data	72%

The large difference between training and testing accuracy suggests possible overfitting.

The following measures can be used:

- Training and testing accuracy.
- Precision and recall.
- F1-score.
- Confusion matrix.
- Cross-validation results.

For churn prediction, recall can be important because failing to identify customers who are likely to leave may cause the company to lose them.

(iii) Apply debugging and regularization techniques

1. L1 regularization

L1 regularization penalizes the absolute values of the coefficients. It can reduce the contribution of unnecessary features and set some coefficients to zero.

2. L2 regularization

L2 regularization penalizes the squared values of coefficients. It discourages excessively large coefficient values.

3. Bayesian priors

Bayesian logistic regression uses prior distributions over the coefficients. A suitable prior can discourage implausibly large values and reduce overfitting.

4. Cross-validation

Evaluate the model across different training and validation splits to check whether its performance is consistent.

5. Feature selection

Remove irrelevant and redundant features to simplify the model.

6. Improve data quality

Correct incorrect labels, handle missing values, and ensure that the training data represent the customers on whom the model will be used.

Conclusion

Overfitting can be reduced by applying suitable Bayesian priors, regularization, feature selection, and cross-validation. These methods help the model generalize better and make more reliable customer churn predictions.

6ANS:

(i) Identify how feature scaling issues affect model performance

Linear regression predicts continuous values using relationships between input features and an output variable.

For example, a sales prediction model may use:

- Advertising budget measured in thousands of rupees.
- Number of outlets measured in individual units.

These features have different numerical ranges.

Different scales can affect the efficiency of optimization algorithms and the numerical

stability of some calculations. They can also make the magnitudes of coefficients difficult to compare directly.

However, different feature scales alone do not necessarily cause poor predictions in ordinary least-squares linear regression. Other issues, such as missing features, noisy data, or an unsuitable model, should also be investigated.

(ii) Analyze the impact of unscaled features on regression coefficients

The linear regression model is:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Where:

- Y = Predicted sales.
- x_1= Advertising budget.
- x_2= Number of outlets.
- β_0= Intercept.
- β_1, β_2= Regression coefficients.

The coefficients depend on the units used to measure the input features.

For example, advertising expenditure measured in rupees will have a different coefficient from the same expenditure measured in thousands of rupees.

When the input units are changed consistently, ordinary least-squares predictions can remain unchanged because the coefficients adjust to the new units.

(iii) Apply optimization techniques

1. Standardization

Standardization transforms features to have a mean of approximately zero and a standard deviation of one.

$$Z = \frac{x-\mu}{\sigma}$$

Where:

- X = Original feature value.
- μ = Mean of the feature.
- σ = Standard deviation.

2. Min-max normalization

Normalization converts feature values into a selected range, commonly between 0 and 1.

$$X' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

3. Improve numerical optimization

Scaling can help gradient-based algorithms converge more efficiently.

4. Prevent data leakage

Fit the scaler using only the training data, then apply the same transformation to validation and testing data.

5. Evaluate the improved model

Compare MAE and RMSE before and after scaling to determine whether predictive performance has improved.

Conclusion

Feature scaling improves numerical conditioning and can make model training more efficient. The company should also investigate data quality, feature selection, and model assumptions to identify the actual cause of poor sales prediction.

7ANS:

(i) Identify the problem caused by class imbalance

Class imbalance occurs when one class has significantly more observations than the other.

For example, a transaction dataset may contain:

- 9,900 legitimate transactions.
- 100 fraudulent transactions.

A model that predicts every transaction as legitimate would achieve 99% accuracy, even though it fails to detect any fraud.

Therefore, accuracy alone cannot adequately measure the performance of a fraud detection model.

(ii) Evaluate model performance using suitable metrics

A confusion matrix helps evaluate the classification results.

Actual / Predicted	**Fraud**	**Legitimate**
Fraud	True Positive (TP)	False Negative (FN)
Legitimate	False Positive (FP)	True Negative (TN)

1. Precision

Precision measures the proportion of transactions predicted as fraud that are actually fraudulent.

$$\text{Precision} = \frac{TP}{TP+FN}$$

2. Recall

Recall measures the proportion of actual fraudulent transactions correctly detected.

$$\text{Recall} = \frac{TP}{TP+FN}$$

3. F1-score

The F1-score combines precision and recall.

$$\text{F1} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Recall is important because low recall means many fraudulent transactions are missed. Precision is also important because excessive false alarms can inconvenience legitimate customers.

(iii) Apply optimization techniques

1. Oversampling

Increase the representation of fraudulent transactions in the training data. Techniques such as SMOTE can generate synthetic minority-class examples when appropriate.

2. Undersampling

Reduce the number of legitimate transactions used during training to make the classes more balanced.

3. Class weighting

Assign a higher penalty to incorrect predictions involving fraudulent transactions.

4. Threshold adjustment

Adjust the classification threshold to improve fraud detection according to the cost of false negatives and false positives.

5. Use suitable evaluation methods

Evaluate the model using precision, recall, F1-score, and the precision-recall curve.

6. Avoid data leakage

Perform oversampling or undersampling only on the training data, not on the complete dataset before splitting it.

Conclusion

Class imbalance can make a fraud detection model favour legitimate transactions. Resampling, class weighting, threshold adjustment, and suitable evaluation metrics can improve the detection of fraudulent transactions.

8ANS:

(i) Identify how correlated features affect Naïve Bayes performance

Naïve Bayes assumes that the input features are conditionally independent given the class.

For example, in an email spam detection system, the following features may be correlated:

- Frequency of promotional words.
- Frequency of discount-related words.
- Number of marketing-related phrases.

When correlated features are treated as independent, the classifier may count similar evidence multiple times. This can distort its estimated class probabilities.

However, correlated features do not always reduce classification accuracy. The effect depends on the dataset and the relationships between the features.

(ii) Analyze feature relationships in the dataset

The following steps can identify correlated features:

1. Calculate correlations between numerical features.
2. Create a correlation matrix or heatmap.
3. Identify features that contain similar information.
4. Examine the importance of each feature.
5. Compare model performance before and after removing redundant features.

Pearson correlation can be used to examine linear relationships between numerical variables. Suitable association measures should be used for categorical variables.

For example, if two features measure almost the same characteristic, retaining both may provide little additional information.

(iii) Apply optimization techniques

1. Feature selection

Remove highly redundant or irrelevant features while retaining those that provide useful information.

2. Dimensionality reduction

Use methods such as Principal Component Analysis (PCA) for suitable numerical datasets to represent information using fewer variables. However, PCA does not guarantee conditional independence between the transformed features.

3. Feature engineering

Combine related variables into a more meaningful feature when appropriate.

4. Compare alternative models

Evaluate Naïve Bayes against other suitable classifiers if feature dependencies substantially affect its performance.

5. Evaluate the results

Use accuracy, precision, recall, and F1-score on unseen data to compare the original and improved models.

Conclusion

Highly correlated features can cause Naïve Bayes to overcount similar evidence and produce distorted probability estimates. Feature selection, dimensionality reduction, and model comparison can help improve classification performance.

9ANS:

(i) Identify the cause of overlapping decision boundaries

Logistic regression is a supervised learning algorithm that estimates the probability of a class using a logistic function.

Overlapping feature distributions occur when observations belonging to different classes have similar input values.

The possible causes are:

1. **Insufficient features:** The selected features may not adequately distinguish the classes.
2. **Nonlinear relationships:** The relationship between the input features and class labels may not be linear.
3. **Noisy data:** Incorrect or inconsistent observations may affect the decision boundary.
4. **Incorrect labels:** Some observations may have been assigned to the wrong class.
5. **Natural class overlap:** The classes may genuinely share similar characteristics.

(ii) Analyze the feature space and classification errors

The logistic regression model is:

$$P(Y=1 \mid X) = \frac{1}{1+e^{-z}}$$

Where:

$$Z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Here:

- x_1, x_2= Input features.
- $\beta_0, \beta_1, \beta_2$= Regression coefficients.

The model estimates the probability that an observation belongs to class 1.

Using a threshold of 0.5, an observation is classified as class 1 when its predicted probability is at least 0.5.

To analyze the problem:

1. Plot the input features to examine class overlap.
2. Create a confusion matrix.
3. Identify false positives and false negatives.
4. Evaluate precision, recall, and the ROC curve.
5. Examine whether important features are missing.

(iii) Improve the model using optimization techniques

1. Feature transformation

Transform the original features to reveal relationships that are not easily represented in their original form.

2. Polynomial features

Add terms such as:

$$x_1^2, \qquad x_2^2, \qquad x_1 x_2$$

These terms allow logistic regression to represent nonlinear decision boundaries in the original feature space.

3. Regularization

L1 or L2 regularization can help control model complexity and reduce overfitting.

4. Feature selection

Add relevant features and remove irrelevant or redundant ones.

5. Threshold adjustment

Change the classification threshold according to the relative costs of false positives and false negatives.

6. Use another suitable model

If the relationship is strongly nonlinear, compare logistic regression with an appropriate nonlinear classifier.

Conclusion

Overlapping classes may limit the performance of a simple logistic regression model. Feature transformation, polynomial features, and suitable regularization can improve classification when the original model does not adequately represent the data.

10ANS:

(i) Identify the signs of underfitting

Underfitting occurs when a machine learning model is too simple to learn the important patterns in the training data.

The main signs are:

1. **Low training accuracy:** The model cannot classify even the training examples correctly.
2. **Low testing accuracy:** The model also performs poorly on unseen data.
3. **High prediction error:** The model fails to capture important relationships.
4. **Limited feature representation:** Important variables or interactions may be missing.
5. **High bias:** The model's assumptions may be too restrictive for the dataset.

(ii) Analyze model bias and feature limitations

Linear Discriminant Analysis (LDA) is a classification technique that models class distributions and produces linear decision boundaries under its standard assumptions.

If the actual classes have nonlinear boundaries, LDA may not represent the data sufficiently well.

For example, two classes may be separated by a curved boundary rather than a straight line.

Underfitting may also occur when:

- Important features are missing.
- The input variables contain insufficient information.
- The model assumptions do not match the data.
- The model cannot represent important feature interactions.

To analyze the problem:

1. Compare training and testing accuracy.
2. Plot the feature distributions for each class.
3. Examine the decision boundary.
4. Identify missing or uninformative features.
5. Check whether the assumptions of the model are appropriate.

(iii) Apply optimization strategies

1. Add relevant features

Include additional variables that help distinguish between classes.

2. Feature transformation

Transform the original variables to capture more complex relationships.

3. Add polynomial features

Include squared terms or interaction terms where appropriate.

4. Increase model flexibility

Consider Quadratic Discriminant Analysis (QDA). Unlike LDA, QDA allows different covariance matrices for different classes and can produce quadratic decision boundaries.

5. Improve data quality

Correct incorrect labels, handle missing values, and remove data-entry errors.

6. Evaluate the improved model

Compare training and testing performance using accuracy, precision, recall, and a confusion matrix.

If both training and testing performance improve, the new model may be capturing the underlying patterns more effectively.

Conclusion

Underfitting occurs when a model cannot represent the complexity of the data. Adding informative features, transforming variables, and selecting a more flexible model can improve classification performance.